

1. Planning & Scope Definition

- Apply **3PDF** Framework to justify updating Yoo & Harman (2012)
- Define **SLR Goals** and **RQs**

2. Search Strategy

- **Databases:** IEEE Xplore, ACM Digital Library, ScienceDirect
- **Time span:** 2013 – May 2025

Output: 499 papers

3. Study Selection

- **Inclusion:** Relevance to TSM, TCS, TCP
- **Exclusion:** language, accessibility, peer-review, length, duplicates

Output: 109 papers

4. Snowballing

- **Backward** and **Forward** snowballing
- Deduplication

Output: 198 papers

5. Classification

- Classification into **SLR, Survey/Review, Primary Study**

6. Quality Appraisal

- Venue Quality (**IDQ**) + DARE-based assessment (**CDQ**)
- **Threshold:** $FDQ = \text{mean}(IDQ, CDQ) \geq 0.6$

Output: 121 high-quality papers

7. Data Extraction

- **RQ1:** publication trends, reproducibility
- **RQ2:** Taxonomy, Algorithms, Metrics
- **RQ3:** Datasets/SUTs

8. Data Synthesis

- Coding, aggregation, comparative analysis, gap identification